TPS Commissioning
Oral Exam Master Packet

Dose calculation algorithms, beam modeling, validation, QA, and clinical traps

Purpose

This packet is written to be the kind of document you can *study from directly* and also *speak from in an oral exam*. It is intentionally organized around questions examiners tend to ask:

- What does TPS commissioning actually mean?
- What data do you input into the TPS?
- How do PB, CCC, AAA, Acuros, and Monte Carlo calculate dose?
- How do kernel-based algorithms account for heterogeneity?
- How do you validate the TPS and what tolerances do you use?
- How do TG-53, TG-106, MPPG 5.a, TRS-430, TG-65, TG-105, TG-157, and TG-329 fit together?
- What do you tweak if the beam model does not agree with measurement?

How to use this packet

Read the blue and green boxes first. Then work through the algorithm sections out loud. Finally, read the *Rapid-fire oral exam answers* until the language feels natural.

Prepared as a self-contained LaTeX study document

No TikZ required; figures are shown as lightweight boxed diagrams

Contents

1 What TPS Commissioning Is

High-yield point

TPS commissioning is not just loading beam data. It is the complete process of configuring, modeling, validating, documenting, and clinically releasing the treatment planning system so that calculated dose agrees with measured dose within accepted clinical tolerances.

Oral-exam answer

What is the goal of TPS commissioning?

The goal is to prove that the TPS can accurately calculate dose for the clinical beams and techniques used in the department. That includes open fields, modified fields, irregular fields, heterogeneous geometries, IMRT/VMAT fields, and clinical end-to-end workflows.

A safe oral answer is:

Commissioning teaches the TPS our machine. Validation proves the TPS learned correctly. Routine QA proves it stayed correct.

Definition

Acceptance testing versus commissioning versus validation

- **Acceptance testing**: verifies that the purchased system, software, hardware, licenses, interfaces, and basic functions were delivered and work as specified.
- **Commissioning**: configures the dose algorithms, beam models, CT-density tables, machine data, and clinical workflows for local use.
- **Validation**: tests the commissioned model using independent data that were not used to tune the model.
- **Routine QA**: verifies that the TPS remains consistent after time, upgrades, workflow changes, and database modifications.

Common trap

The oral-exam trap: "We validated the beam model using the same water-tank data used to tune it."

That is not independent validation. Tuning data can show that the model was fit. Validation data must show that the model predicts new situations correctly.

Clinical workflow

High-level TPS commissioning workflow

1. Acceptance test the TPS hardware, software, licenses, and interfaces.
2. Acquire high-quality beam data.
3. Process beam data minimally and transparently.
4. Commission CT-to-density / material conversion tables.

5. Configure the beam model and dose calculation algorithm.
6. Tune model parameters against measured data.
7. Validate simple open fields.
8. Validate modifiers: wedges, MLCs, bolus, blocks, electron cones, etc.
9. Validate heterogeneity corrections.
10. Validate IMRT/VMAT and dynamic delivery.
11. Perform end-to-end tests from CT simulation through treatment delivery.
12. Document limitations, results, tolerances, and clinical release approval.

1.1 Model-Based versus Measurement-Based Planning

Oral-exam answer

What is the difference between model-based and measurement-based dose calculation?

A **measurement-based** approach relies heavily on measured lookup tables and empirical correction factors. It can work well near the measurement conditions but may extrapolate poorly in complex geometries.

A **model-based** approach uses measured data to tune a physical model of the beam and radiation transport. Model-based systems can predict dose in clinical geometries that were not directly measured, but their accuracy depends on the correctness of the model assumptions and input data.

High-yield point

Memory phrase:

Measurements anchor the model. The model predicts the clinical geometry.

1.2 When Is a TPS Ready for Validation?

Oral-exam answer

How do you determine if the TPS is ready for validation?

The TPS is ready for independent validation only after the internal beam model is stable. That means:

- open-field PDDs and profiles match without unexplained systematic trends,
- output factors agree across field size,
- wedge and modifier data are plausible,
- MLC transmission and DLG-like effects are reasonable,
- CT-density tables are approved,
- absolute dose normalization is configured correctly,
- data transfer pathways work,
- and there are no large unexplained residuals being hidden by smoothing or local overrides.

Common trap

Good oral phrase:

The TPS is ready for validation when the model has stopped needing rescue. If I still need large smoothing, unexplained parameter compensation, or repeated retuning after each test, I am not ready for independent validation.

2 Dose Calculation Algorithms

2.1 Big-Picture Algorithm Comparison

Table 1: Dose calculation algorithm comparison for oral exams.

Algorithm	How it works	Heterogeneity handling	Transport view	Exam-ready strengths / weaknesses
Pencil beam	Splits field into narrow beamlets and applies simplified kernels	Usually equivalent path length or density scaling; weak lateral electron transport	Approximate beamlet transport	Very fast, but weakest in lung, air cavities, interfaces, and small fields.
Convolution / Superposition	Computes energy release and spreads dose using kernels	Density scaling / radiological scaling of dose deposition kernels	Kernel-based transport approximation	Major improvement over PB; clinically robust for many photon cases.
Collapsed cone	Efficient convolution/superposition by transporting energy along discrete cones	Kernel deformation and density scaling along cone directions	Discrete kernel transport	Good balance of speed and accuracy; still approximate near severe interfaces.
AAA	Anisotropic analytical convolution/superposition model with primary, extra-focal, and electron contamination components	Anisotropic density-aware kernel scaling	Analytical model-based approximation	Fast and clinically useful; less accurate than Acuros/MC in severe heterogeneity.
Acuros XB	Deterministic solution of the linear Boltzmann transport equation	Explicit material-based transport	LBTE solver	Excellent heterogeneity accuracy; depends on material assignment and CT calibration.
Monte Carlo	Randomly simulates particle histories	Natural material-specific transport	Stochastic particle transport	Gold standard physics; slower and subject to statistical uncertainty.

Oral-exam answer

What is the hierarchy of dose calculation algorithms?

A practical hierarchy is:

$$\text{Pencil Beam} \rightarrow \text{Convolution/Superposition} \rightarrow \text{Collapsed Cone / AAA} \rightarrow \text{LBTE / Acuros} \rightarrow \text{Monte Carlo.}$$

As you move to the right, the radiation transport physics generally becomes more realistic, especially in heterogeneity, but computation becomes more complex.

2.2 Pencil Beam

Definition

Pencil beam

A pencil beam algorithm divides the treatment beam into small beamlets and applies a simplified dose deposition kernel to each beamlet. The final dose is obtained by summing the beamlet contributions.

Oral-exam answer

Why do pencil beam algorithms fail in lung?

Pencil beam algorithms primarily correct for attenuation along the beam direction. They do not accurately model lateral electron transport. In lung, secondary electrons travel farther because the density is low, so charged particle equilibrium is lost. That causes PB algorithms to perform poorly in small lung fields, interfaces, and low-density regions.

High-yield point

One-line exam answer:

Pencil beam knows where the photons went better than it knows where the electrons went.

2.3 Convolution / Superposition

Definition

Convolution/superposition

Convolution/superposition algorithms calculate dose by combining the energy released by photon interactions with a dose deposition kernel that describes how secondary charged particles spread dose around the interaction site.

Key physics

Dose calculation concept

$$D(\vec{r}) = \int \mathrm{TERMA}(\vec{r}')K(\vec{r} - \vec{r}', \rho)\, d\vec{r}' \tag{1}$$

where:

- TERMA represents total energy released per unit mass,
- K is the dose deposition kernel,
- ρ represents density or material dependence.

Oral-exam answer

How do kernel-based algorithms account for inhomogeneity?

Kernel-based algorithms account for inhomogeneity by scaling, stretching, compressing, or tilting the dose deposition kernel based on local density or radiological distance. In low density, the deposition cloud spreads farther. In high density, it contracts. This is why convolution/superposition handles heterogeneity better than simple pencil beam methods.

2.4 Collapsed Cone Convolution

Definition

Collapsed cone convolution

Collapsed cone convolution is an efficient form of convolution/superposition where the 3D dose deposition kernel is collapsed into a finite number of cones or rays. Energy is transported along these cones through the patient geometry.

Oral-exam answer

Explain collapsed cone convolution algorithm.

First, the algorithm ray traces the primary photon fluence through the patient and computes the energy released in each voxel. Then it transports that released energy along a finite set of cone directions using density-scaled kernels. This approximates the 3D spread of secondary electrons and scattered photons while remaining fast enough for clinical use.

High-yield point

High-yield phrase:

Collapsed cone improves heterogeneity handling because it deforms the deposition cloud using the medium density rather than using only equivalent path length.

2.5 AAA

Definition

AAA

The Anisotropic Analytical Algorithm is an analytical convolution/superposition algorithm used in Eclipse. It models dose using primary photon fluence, extra-focal photon scatter, and contaminant electron components.

Oral-exam answer

What are the strengths and weaknesses of AAA?

AAA is fast, clinically practical, and much better than older pencil beam algorithms in heterogeneous media. However, it is still approximate. It can struggle in very small lung fields, air cavities, severe density interfaces, and charged-particle disequilibrium situations.

Common trap

Do not say:

"AAA is Monte Carlo-like."

AAA uses Monte-Carlo-derived data in its modeling, but clinically it is still an analytical model-based algorithm. It does not explicitly simulate particle histories and does not solve the LBTE directly.

2.6 Acuros XB and the LBTE

Definition

Linear Boltzmann Transport Equation

The LBTE describes radiation transport by accounting for particle streaming, attenuation, scattering, and source production.

Key physics

Conceptual LBTE form

$$\underbrace{\Omega \cdot \nabla \Psi}_{\text{streaming}} + \underbrace{\sigma_t \Psi}_{\text{collision / attenuation}} = \underbrace{Q}_{\text{source}} \tag{2}$$

Oral-exam answer

How is Acuros different from AAA?

AAA uses analytical density-scaled kernels. Acuros XB solves a discretized form of the LBTE. Therefore, Acuros models photon and electron transport more explicitly in assigned materials. This makes Acuros more accurate in heterogeneity, bone, lung, and air-interface situations.

Oral-exam answer

How is Acuros different from Monte Carlo?

Both model radiation transport physics. Monte Carlo samples particle histories randomly and has statistical noise. Acuros solves the transport equation deterministically on a grid, so it has discretization error but not stochastic noise.

2.7 Monte Carlo

Definition

Monte Carlo

Monte Carlo dose calculation simulates particle transport using random sampling of interaction probabilities. A particle history includes the primary particle and all secondary particles it creates until they are absorbed, escape, or fall below transport thresholds.

Key physics

Statistical uncertainty

$$\sigma \propto \frac{1}{\sqrt{N}} \tag{3}$$

where N is the number of particle histories.

Oral-exam answer

What are the main Monte Carlo commissioning issues?

Important issues include:

- source model accuracy,
- CT-to-material conversion,
- particle transport cutoffs,
- variance reduction,
- statistical uncertainty,
- latent variance from phase-space files,
- dose-to-medium versus dose-to-water reporting.

Common trap

Common trap:

Low-noise Monte Carlo does not automatically mean correct Monte Carlo. A smooth MC dose distribution can still be wrong if the source model, CT material map, or transport settings are wrong.

3 Dose Reporting: Dose-to-Water, Dose-to-Medium, and TG-329

Definition

Dose reporting convention

Dose depends on both where energy is deposited and what material radiation is transported through. TPS algorithms may report dose-to-water, dose-to-medium, or an approximation related to dose-to-tissue.

Key physics

Dose-to-medium converted to dose-to-water

$$D_w^m = D_m^m \times \frac{\left(\frac{S}{\rho}\right)^{\text{unres}}_{\text{water}}}{\left(\frac{S}{\rho}\right)^{\text{unres}}_{\text{medium}}} \tag{4}$$

Oral-exam answer

What is the difference between dose-to-water and dose-to-medium?

Dose-to-water reports the dose as if energy were deposited in water. Dose-to-medium reports the dose deposited in the actual material of the voxel. Dose-to-water is historically connected to water calibration and clinical outcome data. Dose-to-medium is more physically local for material-based transport algorithms such as Monte Carlo and Acuros.

Oral-exam answer

What is the TG-329 issue?

TG-329 addresses reference dose specification and the fact that modern TPS algorithms may not all report the same physical quantity. The key clinical point is: know what your TPS reports before applying any manual scaling between water, tissue, or muscle. Otherwise, you can double-count or misapply a correction.

Common trap

Oral exam trap:

Do not blindly say "the TPS reports dose-to-water." Some algorithms can report dose-to-medium or dose-to-water depending on settings. Always say: *I need to know the algorithm and reporting mode.*

4 What Data Go Into the TPS?

4.1 Photon Beam Data

High-yield point

A photon beam model is only as good as the beam data used to build it. TG-106-style commissioning emphasizes high-quality, minimally processed measured data.

Table 2: Common photon beam data inputs for TPS commissioning.

Data type	Purpose
PDDs / TMRs / TPRs	Define depth-dose behavior and effective beam energy.
Profiles at multiple depths	Define flatness, symmetry, horns, penumbra, and off-axis behavior.
Output factors	Define field-size dependence and small-field behavior.
MLC transmission / leakage	Defines leakage and dynamic delivery dose effects.
DLG / leaf-tip behavior	Defines rounded leaf-end dosimetry and small-segment output behavior.
Wedge factors / wedge profiles	Defines physical or dynamic wedge attenuation, hardening, and profile shape.
Tray / block / accessory factors	Defines accessory attenuation.
Absolute dose calibration	Links calculated dose to MU output under reference conditions.

4.2 Electron Beam Data

Table 3: Common electron beam data inputs for TPS commissioning.

Data type	Purpose
Electron PDDs	Defines range, R_{90}, R_{80}, R_{50}, and depth-dose behavior.
Profiles at multiple depths	Defines flatness, penumbra, and scatter behavior.
Cone factors	Defines applicator output.
Cutout factors	Defines insert-dependent output behavior.
Effective SSD / virtual source data	Defines extended SSD behavior.
Irregular insert measurements	Validates clinical cutout modeling.

4.3 CT-Density Table

Oral-exam answer

How do you generate a CT-density table?

You scan a calibrated electron-density phantom using the clinical CT protocol. For each insert, measure the mean HU and map it to relative electron density or mass density. Then load that curve into the TPS and verify it using a phantom scan. The table should be scanner-specific and protocol-specific when HU behavior changes significantly with scanner, kVp, reconstruction, or artifact correction.

High-yield point

How many CT-density tables do you need?

A safe answer is:

At least one per CT scanner used for planning, and additional tables for clinically used kVp or reconstruction protocols if they produce meaningfully different HU-density relationships.

4.4 Machine Metadata and Geometry

Oral-exam answer

What non-dose data does the TPS need?

The TPS needs machine geometry and metadata, including:

- SAD and SSD conventions,
- jaw limits,
- MLC leaf widths and geometry,
- couch and immobilization modeling assumptions,
- wedge geometry,
- electron cone geometry,
- treatment head limits,
- delivery technique definitions,
- coordinate system and DICOM conventions.

5 Water Tank Commissioning and Data Processing

5.1 Commissioning a Water Tank

Clinical workflow

Water tank setup checklist

1. Confirm tank size is sufficient for the largest fields and scan lengths.
2. Level the tank.
3. Verify tank motion and scanner orthogonality.

4. Align the detector to the beam central axis.
5. Establish water surface / zero depth carefully.
6. Verify SSD independently.
7. Select appropriate detector for field size and gradient.
8. Confirm scan direction and coordinate labels.
9. Check beam stability during acquisition.
10. Review scans immediately for noise, shifts, artifacts, and missing tails.

Common trap

Common water-tank mistakes

- wrong zero depth,
- tank not level,
- detector too large for penumbra or small fields,
- wrong effective point of measurement,
- water ripple,
- cable effects,
- insufficient scan length outside the field,
- over-smoothing bad data.

5.2 Data Processing

Oral-exam answer

What data processing operations might be used?

Common processing operations include smoothing, centering, mirroring, interpolation, normalization, deconvolution, and scan stitching. These operations can be useful, but they must be documented and justified.

High-yield point

Memory phrase:

Bad data gets prettier before it gets dangerous.

6 Beam Modeling Parameters and Troubleshooting

6.1 What Can You Tweak in a Beam Model?

Table 4: Data processing operations and what they can change.

Operation	What it changes	Caution
Smoothing	Reduces noise	Can blunt penumbra, alter flatness, and hide artifacts.
Centering	Aligns scan to CAX	Can falsely hide setup error or shift penumbra.
Mirroring	Forces symmetry	Should not be used to hide real beam asymmetry.
Interpolation	Changes point spacing	Does not create real measured information.
Normalization	Sets relative scale	Wrong normalization can make PDD/profile comparisons misleading.
Deconvolution	Corrects detector-volume blurring	Powerful but must be validated carefully.

High-yield point

Beam-model parameters commonly represent:

- effective beam energy / spectrum,
- source size,
- primary fluence,
- flattening filter,
- extra-focal radiation,
- electron contamination,
- jaw transmission,
- MLC transmission,
- DLG / rounded leaf-tip behavior,
- tongue-and-groove,
- wedge attenuation and beam hardening,
- output-factor scaling.

Oral-exam answer

If the penumbra does not line up, what do you check?
I check source size, detector size, scan step size, MLC/jaw geometry, leaf-tip model, and whether the measured profile was centered correctly. Penumbra disagreement is not usually fixed by changing beam energy.

Oral-exam answer

If shallow PDD does not agree, what do you check?
I check electron contamination modeling, detector choice, zero depth, effective point of measurement, surface setup, and buildup measurement quality. Shallow-dose errors often come from measurement and contamination modeling rather than the deep photon spectrum alone.

Table 5: Beam model troubleshooting table.

Observed issue	Parameter to inspect	Reasoning
PDD too penetrating at depth	Energy spectrum / effective energy	Deep PDD is strongly energy dependent.
Shallow PDD mismatch	Electron contamination, surface setup, detector correction	Buildup is sensitive to contaminant electrons and zero depth.
Penumbra too wide	Source size too large, detector-volume blurring	Larger effective source broadens geometric penumbra.
Penumbra too sharp	Source size too small, leaf-tip model wrong	Too-small source artificially sharpens profiles.
Large-field horns wrong	Flattening filter / off-axis softening	Horns reflect beam profile and flattening filter modeling.
Output factors wrong for small fields	Source size, MLC geometry, small-field detector data	Small fields are sensitive to source occlusion and detector effects.
MLC sweep dose wrong	MLC transmission, DLG, tongue-and-groove	Dynamic MLC dose depends on leaf-end and leakage modeling.
Wedge factor wrong	Wedge model, depth convention, field-size dependence	Physical wedge factors depend on depth and field size.
Heterogeneity validation fails	CT-density table, algorithm limitation, material assignment	Water-only agreement does not guarantee heterogeneity accuracy.

6.2 Wedge Factor

Key physics

Wedge factor

$$WF = \frac{D_{\text{wedged}}}{D_{\text{open}}} \tag{5}$$

measured at the same geometry, depth, field size, energy, and normalization convention.

Oral-exam answer

How do you determine wedge factor?

Measure dose with the wedge and divide by dose without the wedge for the same field size, depth, energy, and setup. For physical wedges, wedge factor depends on field size and depth because the wedge attenuates and hardens the beam. For dynamic or virtual wedges, the dependence is vendor-specific and often depends strongly on moving-jaw direction.

Common trap

Do not measure one wedge factor and use it everywhere.

For physical wedges, wedge factor is not purely a constant. It can depend on wedge angle, field size, depth, energy, and off-axis location.

7 Validation Strategy and Tolerances

7.1 Validation Categories

High-yield point

A strong TPS validation program tests:

1. basic open-field photon dose calculation,

2. modifiers and irregular fields,
3. heterogeneity corrections,
4. small fields and MLC-shaped fields,
5. IMRT/VMAT,
6. electron beams,
7. CT-to-density behavior,
8. end-to-end workflow.

7.2 TG-53 Table 4-4 Summary

Table 6: Suggested acceptability criteria for external beam dose calculations, adapted from TG-53 Table 4-4 and reproduced in TRS-430. Values are example achievable criteria, not universal requirements.

Situation	Abs. dose at norm pt (%)	Central axis (%)	Inner beam (%)	Penumbra (mm)	Outer beam (%)	Buildup region (%)
Homogeneous phantoms						
Square fields	0.5	1	1.5	2	2	20
Rectangular fields	0.5	1.5	2	2	2	20
Asymmetric fields	1	2	3	2	3	20
Blocked fields	1	2	3	2	5	50
MLC-shaped fields	1	2	3	3	5	20
Wedged fields	2	2	5	3	5	50
External surface variations	0.5	1	3	2	5	20
SSD variations	1	1	1.5	2	2	40
Inhomogeneous phantoms* **						
Slab inhomogeneities	3	3	5	5	5	–
3-D inhomogeneities	5	5	7	7	7	–

Notes: Percentages are quoted as a percent of the central-ray normalization dose. Absolute-dose values at the normalization point are relative to a standard beam calibration point. Inhomogeneous-phantom criteria exclude regions of electronic disequilibrium.

Oral-exam answer

What validation tolerances should you know for oral exams?

For simple homogeneous geometry, remember:

- reference point absolute dose: about 0.5%,
- central axis: about 1%,
- inner beam: about 1.5%,
- penumbra: about 2 mm.

Then explain that tolerances loosen for wedges, MLC-shaped fields, buildup, and heterogeneity.

7.3 How to Check Agreement

Clinical workflow

Agreement evaluation hierarchy

1. Point dose for absolute normalization.
2. PDD and profiles for depth and shape.
3. Region-specific criteria for inner beam, penumbra, tail, buildup, and heterogeneity.
4. Gamma or distance-to-agreement for planar/volumetric comparisons.
5. End-to-end testing for workflow and transfer.

Common trap

Why one number is dangerous

A single 3%/3 mm gamma passing rate can hide a systematic error in one clinically important region. Always inspect where failures occur.

8 Heterogeneity and TG-65

Oral-exam answer

How does the TPS handle inhomogeneities?

Simple methods use equivalent path length or radiological depth. Kernel-based algorithms scale the dose deposition kernel according to density. AAA uses anisotropic density-aware kernels. Acuros and Monte Carlo perform more explicit material-based transport.

Definition

Radiological depth

$$d_{\text{rad}} = \int \rho_e(s)\, ds \tag{6}$$

Radiological depth is the density-weighted path length through the patient.

High-yield point

Best oral phrase:

Large-field heterogeneity is often a path-length problem. Small-field heterogeneity and interfaces are charged-particle-equilibrium problems.

Oral-exam answer

What heterogeneity tests should you do?

Use slab lung, slab bone, oblique incidence, 3D heterogeneous phantoms, thorax-like geometries, small fields in lung-equivalent material, and clinical IMRT/VMAT heterogeneous validation cases.

Common trap

TG-53 heterogeneity caveat

The inhomogeneous phantom tolerances exclude regions of electronic disequilibrium. That matters because directly at interfaces, the physics is harder and measurement uncertainty is higher.

9 IMRT and VMAT Commissioning

9.1 What Changes for IMRT?

Oral-exam answer

What is different about IMRT commissioning?

IMRT commissioning stresses small fields, MLC transmission, DLG, tongue-and-groove, leaf sequencing, output factors, and highly modulated dose delivery. Water-tank open-field agreement is necessary but not sufficient.

9.2 What Changes for VMAT?

Oral-exam answer

What do you pay closer attention to for VMAT?

VMAT requires validation of:

- dynamic MLC motion,
- dose rate modulation,
- gantry speed,
- control point interpolation,
- final calculation versus optimization dose engine,
- plan transfer and delivery timing.

High-yield point

One-line exam answer:

VMAT is not just IMRT delivered faster. It couples MLC motion, gantry motion, and dose-rate modulation.

9.3 IMRT/VMAT Validation Tests

Clinical workflow

Useful IMRT/VMAT validation tests

- TG-119 style benchmark plans,
- small field output tests,
- sweeping gap tests,

- MLC transmission and DLG tests,
- highly modulated head-and-neck cases,
- lung heterogeneity IMRT/VMAT cases,
- patient-specific QA with ion chamber and array/film,
- end-to-end CT-to-delivery test.

10 TPS QA Frequency and Clinical Maintenance

Oral-exam answer

How often do you QA your TPS?

TPS QA is performed periodically and after any major change. That includes:

- software upgrade,
- algorithm upgrade,
- beam model change,
- CT scanner change,
- CT protocol change,
- R&V interface change,
- delivery-system change,
- database migration.

Routine QA typically includes reference plan recalculation, CT-density checks, image transfer checks, MU/time verification, and electronic plan transfer checks.

Table 7: Routine TPS QA items.

Item	**Typical trigger/frequency**	**Purpose**
Reference plan recalculation	Annual / after upgrade	Compare new calculation to commissioning baseline.
CT-density constancy	Periodic / after CT change	Verify HU-to-density stability.
DICOM import/export	Periodic / after interface change	Verify image, structure, and plan integrity.
Plan transfer to R&V	After upgrades / periodic	Verify MLC, jaws, gantry, couch, dose, MU, wedges, accessories.
Backup and restore	Periodic	Confirm database recoverability.
User access / security	Periodic	Ensure correct permissions and change control.

10.1 Connectivity Testing

Clinical workflow

End-to-end connectivity chain

$$\text{CT} \rightarrow \text{TPS} \rightarrow \text{R\&V} \rightarrow \text{Linac / MLC / Couch / Accessories} \rightarrow \text{IGRT}$$

Check:

- patient orientation,
- image scaling,
- structure transfer,
- beam parameters,
- MLC positions,
- wedge/accessory transfer,
- DRR generation,
- couch modeling,
- dose/MU transfer,
- treatment delivery agreement.

Common trap

Important:
A plan can be dosimetrically perfect in the TPS and still unsafe if transfer to R&V or machine delivery is wrong.

11 Acceptance Testing for a New TPS

Oral-exam answer

What would you check in acceptance testing for a new TPS?
I would verify:

- correct software version,
- correct licenses and algorithms,
- DICOM import/export,
- CT image orientation and scaling,
- structure set import,
- beam model availability,
- dose calculation execution,
- plan transfer to R&V,

- DRR export,
- user access/security,
- backup and restore,
- vendor documentation and known limitations.

High-yield point

Acceptance testing is not full commissioning.

Acceptance testing tells you the system works as delivered. Commissioning tells you it works accurately for your clinic.

12 Optimization Algorithms

Definition

Inverse planning optimization

Inverse planning means the planner defines goals, priorities, and constraints, and the optimizer searches for beam intensities or machine parameters that best satisfy those objectives.

Oral-exam answer

What optimization algorithms are used in inverse planning?

Common concepts include:

- gradient-based optimization,
- simulated annealing,
- fluence-map optimization,
- direct aperture optimization,
- direct machine parameter optimization,
- multi-criteria optimization,
- biological optimization using TCP/NTCP/EUD-style objectives.

Oral-exam answer

What is the difference between fluence optimization and direct machine parameter optimization?

Fluence optimization first finds ideal beamlet intensities, then converts them into deliverable segments or control points. Direct machine parameter optimization optimizes deliverable MLC positions, weights, dose rates, or gantry parameters more directly. VMAT relies heavily on deliverability-aware optimization.

Common trap

Common trap:

A beautiful optimized fluence map is not clinically useful unless it can be delivered accurately by the machine.

13 What to Include in a Commissioning Report

High-yield point

A strong TPS commissioning report should include:

- scope and clinical beam list,
- software version and algorithm version,
- machine and beam configuration,
- measurement equipment and calibration traceability,
- water-tank setup and detector selection,
- measured beam data,
- data processing steps,
- CT-density table method,
- beam model parameters and tuning decisions,
- validation tests and results,
- tolerance criteria and justification,
- heterogeneity validation,
- IMRT/VMAT validation,
- end-to-end transfer testing,
- known limitations,
- routine QA baselines,
- approval signatures and release date.

Common trap

Best documentation rule:

If someone cannot reconstruct what you tested, what you accepted, and what limitations remain, the commissioning report is incomplete.

14 Rapid-fire Oral Exam Answers

14.1 The shortest answers you should be able to give cleanly

Oral-exam answer

TPS commissioning

TPS commissioning is the process of configuring the treatment planning system so its dose calculation, beam model, CT-density mapping, MU logic, and data transfer accurately represent the local clinic.

Oral-exam answer

Acceptance versus commissioning

Acceptance testing verifies that the system delivered by the vendor works as specified. Commissioning verifies that the TPS accurately models the local machines, CT scanners, clinical workflows, and dose calculation requirements.

Oral-exam answer

Validation

Validation is independent testing of the commissioned TPS. It should use measurements or benchmark cases that were not used to tune the beam model.

Oral-exam answer

Pencil beam

Pencil beam is fast but limited. It mostly handles heterogeneity using path-length or density corrections and does not accurately model lateral electron transport, so it performs poorly in lung, small fields, and interfaces.

Oral-exam answer

Convolution/superposition

Convolution/superposition computes energy released in the patient and spreads it with dose deposition kernels. It handles heterogeneity better than pencil beam because the kernel is scaled or deformed by density.

Oral-exam answer

Collapsed cone

Collapsed cone is an efficient convolution/superposition method that transports energy along a finite number of cone directions. It balances accuracy and speed and is widely used clinically.

Oral-exam answer

AAA

AAA is an Eclipse analytical convolution/superposition algorithm using primary photon, extra-focal photon, and electron contamination source components. It is fast and clinically useful but still approximate in severe heterogeneity.

Oral-exam answer

Acuros XB

Acuros XB solves the linear Boltzmann transport equation deterministically. It models material-based transport more explicitly than AAA and is generally more accurate in heterogeneity.

Oral-exam answer

Monte Carlo

Monte Carlo simulates particle histories using random sampling of interaction physics. It is the gold standard for radiation transport but requires control of source modeling, CT material assignment, statistical uncertainty, and transport cutoffs.

Oral-exam answer

Dose-to-water versus dose-to-medium

Dose-to-water connects naturally to water calibration and historic clinical experience. Dose-to-medium is more physically local to the actual voxel material and is often native to Monte Carlo or material-based algorithms.

Oral-exam answer

What data do you input into the TPS?

You input measured PDDs, profiles, output factors, wedge/accessory data, MLC transmission and DLG-related data, electron cone/cutout data, CT-density tables, machine geometry, and absolute dose normalization information.

Oral-exam answer

How do you determine wedge factor?

Measure wedged dose divided by open-field dose at the same energy, field size, depth, and geometry. For physical wedges, measure enough depths and field sizes because wedge factor is not purely constant.

Oral-exam answer

When is the TPS ready for validation?

When open-field PDDs, profiles, output factors, modifiers, MLC data, CT-density tables, and MU normalization are internally consistent and no large unexplained residuals remain.

Oral-exam answer

Validation tolerances

For simple homogeneous fields, remember about 0.5% at the reference point, 1% central axis, 1.5% inner beam, and 2 mm penumbra. Loosen criteria for wedges, MLC-shaped fields, buildup, and heterogeneity.

Oral-exam answer

How does the TPS handle heterogeneity?

Simple algorithms use path-length corrections. Kernel-based methods scale the dose deposition kernel. AAA uses anisotropic density-aware kernels. Acuros and Monte Carlo perform more explicit material-based transport.

Oral-exam answer

CT-density table

Scan a density phantom with the clinical protocol, measure HU for each insert, map HU to electron or mass density, load the curve into TPS, and verify scanner/protocol consistency.

Oral-exam answer

Water tank commissioning

Verify tank level, detector alignment, scanner motion, zero depth, SSD, detector choice, scan length, coordinate labels, and beam stability before trusting the data.

Oral-exam answer

Data processing

Use minimal processing. Smoothing, centering, mirroring, interpolation, and deconvolution can be helpful, but each can also distort the true beam data if used carelessly.

Oral-exam answer

If penumbra does not match

Check source size, detector size, scan step size, MLC/jaw geometry, leaf-tip model, and scan centering. Do not first blame beam energy.

Oral-exam answer

If shallow PDD does not match

Check electron contamination, detector choice, zero depth, effective point of measurement, and buildup measurement quality.

Oral-exam answer

IMRT commissioning

IMRT commissioning requires validation of small fields, MLC transmission, DLG, tongue-and-groove, leaf sequencing, output factors, and highly modulated dose distributions.

Oral-exam answer

VMAT commissioning

VMAT adds gantry motion, dose rate modulation, dynamic MLC motion, and control point interpolation. Therefore, VMAT commissioning must verify the dynamic delivery chain, not just static beam data.

Oral-exam answer

TPS QA frequency

TPS QA is periodic and event-driven. Recheck after software upgrades, beam-model changes, CT protocol changes, R&V interface changes, or delivery-system modifications.

Oral-exam answer

Connectivity testing

Verify the full chain: CT to TPS, TPS to R&V, R&V to linac/MLC/couch/accessories, and IGRT reference data. Dose calculation accuracy is not enough if transfer is wrong.

Oral-exam answer

Optimization algorithms

Inverse planning optimizes beam fluence or machine parameters based on objectives and constraints. VMAT requires deliverability-aware optimization because gantry speed, dose rate, and MLC motion are coupled.

15 Final High-Yield Checklist

High-yield point

If you can say the following clearly, you are in excellent shape for the TPS commissioning oral exam:

1. I can distinguish acceptance testing, commissioning, validation, and routine QA.
2. I know what data are required for photon and electron beam modeling.
3. I can explain PB, convolution/superposition, CCC, AAA, Acuros, and Monte Carlo.
4. I know how kernel-based algorithms augment the deposition cloud to account for heterogeneity.
5. I understand dose-to-water versus dose-to-medium and why TG-329 matters.
6. I can explain how to generate a CT-density table.
7. I know the classic TG-53 validation tolerances.
8. I can describe heterogeneity validation and TG-65-style concerns.
9. I can troubleshoot model disagreement using beam-model parameters.
10. I know what changes for IMRT and VMAT commissioning.
11. I know how often to QA the TPS and what triggers recommissioning.
12. I can describe end-to-end connectivity testing.

Common trap

The single best oral-exam closing line

A TPS is only as accurate as its beam model, commissioning measurements, CT-density calibration, and clinical workflow validation.

References

[1] Fraass, B., et al. “Quality assurance for clinical radiotherapy treatment planning: Report of AAPM Radiation Therapy Committee Task Group 53.” *Medical Physics*, 25, 1773–1829, 1998.

[2] AAPM Report No. 85. “Tissue Inhomogeneity Corrections for Megavoltage Photon Beams.” Task Group 65 of the AAPM Radiation Therapy Committee, 2004.

[3] Chetty, I. J., et al. “Report of the AAPM Task Group No. 105: Issues associated with clinical implementation of Monte Carlo-based photon and electron external beam treatment planning.” *Medical Physics*, 34, 4818–4853, 2007.

[4] Das, I. J., et al. “Accelerator beam data commissioning equipment and procedures: Report of AAPM TG-106.” *Medical Physics*, 35, 4186–4215, 2008.

[5] Ezzell, G. A., et al. “IMRT commissioning: Multiple institution planning and dosimetry comparisons, a report from AAPM Task Group 119.” *Medical Physics*, 36, 5359–5373, 2009.

[6] Ma, C.-M., et al. “Beam modeling and beam model commissioning for Monte Carlo dose calculation-based radiation therapy treatment planning: Report of AAPM TG-157.” *Medical Physics*, 47, e1–e18, 2020.

[7] Miften, M., et al. “Tolerance limits and methodologies for IMRT measurement-based verification QA: Recommendations of AAPM TG-218.” *Medical Physics*, 45, e53–e83, 2018.

[8] Miften, M., et al. “Reference dose specification for dose calculations: Report of AAPM Task Group 329.” *Medical Physics*, 2020s.

[9] Smilowitz, J. B., et al. “AAPM Medical Physics Practice Guideline 5.a.: Commissioning and QA of Treatment Planning Dose Calculations — Megavoltage Photon and Electron Beams.” *Journal of Applied Clinical Medical Physics*, 16, 14–34, 2015.

[10] International Atomic Energy Agency. *Commissioning and Quality Assurance of Computerized Planning Systems for Radiation Treatment of Cancer*. Technical Reports Series No. 430, IAEA, Vienna, 2004.

[11] International Atomic Energy Agency. *Specification and Acceptance Testing of Radiotherapy Treatment Planning Systems*. IAEA-TECDOC-1540, Vienna, 2007.

[12] International Atomic Energy Agency. *Commissioning of Radiotherapy Treatment Planning Systems: Testing for Typical External Beam Treatment Techniques*. IAEA-TECDOC-1583, Vienna, 2008.

[13] Podgorsak, E. B. *Radiation Oncology Physics: A Handbook for Teachers and Students*. International Atomic Energy Agency, Vienna, 2005.

[14] Philips Medical Systems. *Pinnacle Treatment Planning System Physics Reference Guide*.

[15] RaySearch Laboratories. *RayStation Physics Reference Manual / White Papers*.

[16] Varian Medical Systems. *Eclipse Algorithms Reference Guide / Physics Reference Guide*.